

EXPLORATORY DATA ANALYSIS REPORT

Decodelabs Data Analytics Internship – Week 2 Assignment

Project Title

Exploratory Data Analysis of Sales Transaction Data Using Python

1. Introduction

Exploratory Data Analysis (EDA) is a critical stage in the data analytics process that involves examining datasets to understand their structure, identify patterns, detect anomalies, and generate insights that support decision-making.

For this assignment, Python was used to perform exploratory analysis on a sales transaction dataset. The analysis focused on understanding the distribution of sales values, identifying high-value transactions, detecting potential outliers, and examining product sales trends over time.

2. Objectives

The primary objectives of this analysis were to:

1. Explore the structure and quality of the dataset.
2. Calculate descriptive statistics to understand data characteristics.
3. Analyze the distribution of sales values.
4. Identify potential outliers and unusually large transactions.
5. Determine which products contribute most frequently to high-value sales.
6. Examine sales trends over time for top-performing products.
7. Generate actionable insights from the dataset.

3. Tools and Technologies

The analysis was conducted using:

- Python, Pandas, NumPy, Matplotlib, and Seaborn.

These tools were used for data manipulation, statistical analysis, and visualization.

4. Dataset Exploration

The dataset was imported into a Pandas DataFrame and subjected to an initial assessment.

The following exploratory checks were performed:

- **Dataset Structure:** The `.info()` function was used to examine the number of records, variables, data types, and memory usage
- **Descriptive Statistics:** The `.describe()` function was used to calculate summary statistics such as count, mean, median, standard deviation, minimum value, and quartiles.
- **Missing Values Assessment:** Missing values were identified using: `df.isnull().sum()`
- **Cardinality Analysis:** Unique values within each variable were examined using `df.nunique()`.

5. Distribution Analysis of Total Sales

To understand how transaction values were distributed, a histogram with Kernel Density Estimation (KDE) was created for the **TotalPrice** variable.

The analysis aimed to determine:

- Whether transaction values were normally distributed.
- The concentration of low- and high-value transactions.
- The presence of skewness.

The distribution showed variation in transaction values, suggesting that certain purchases generated substantially higher revenue than others.

Understanding this distribution provides a foundation for identifying purchasing behaviour patterns and revenue concentration.

6. Outlier Detection

A boxplot was used to identify unusually large sales transactions within the TotalPrice variable.

Method: `sns.boxplot(y=df['TotalPrice'])`

The presence of outliers indicates that a small number of transactions may contribute disproportionately to overall revenue.

Such transactions can significantly influence average sales values and should be investigated further.

8. High-Value Transaction Analysis

The 95th percentile of TotalPrice was calculated to identify exceptionally large transactions.

Method: `percentile_95 = df['TotalPrice'].quantile(0.95)`

Transactions above this threshold were classified as high-value transactions.

This approach allows analysts to focus on the top 5% of revenue-generating orders.

9. High-Value Product Analysis

Products associated with high-value transactions were identified using frequency analysis.

Method: `value_counts()`

The analysis highlighted products that appeared most frequently within the top 5% of sales transactions.

12. Conclusion

The exploratory data analysis successfully achieved its objectives by uncovering patterns, trends, and outliers within the sales dataset.

Using descriptive statistics, distribution analysis, outlier detection, and trend analysis, the study provided valuable insight into transaction behaviour and product performance.

The findings establish a strong analytical foundation for future stages of the data analytics process, including predictive modelling, customer segmentation, sales forecasting, and dashboard development.

Through this exercise, key competencies in data exploration, statistical analysis, data visualization, and analytical thinking were demonstrated using Python and industry-standard data science libraries.